Requirements Gathering

Project Title: Global Energy Shock 2026: Oil, Trade & Economic Exposure Analysis

Program: Digital Egypt Pioneers Initiative (DEPI), Data Analytics Track, Graduation Project

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1. Stakeholder Analysis

The platform serves three institutional stakeholder groups. Each uses the same integrated data to answer a different decision question.

Stakeholder	Organizational Role	Decision-Support Needs	Primary Platform Outputs
Macroeconomic Policy Analysts	Analysts at ministries of finance, central banks and development institutions who assess national vulnerability to external shocks.	Rank economies by crude import burden (% of GDP) and export dependence (% of GDP). Separate importers from exporters. Judge how much of a price shock is amplified by a weak currency. Compare countries within the same income group or region.	Exposure ranking, importer/exporter classification, FX-adjusted import bill, income-group comparison
Global Supply Chain Risk Managers	Risk and procurement leads at multinational firms with operations or suppliers across many countries.	Identify operating markets where an oil price spike would raise the macro-economic stress that affects costs, demand and currency stability. Monitor how exposure shifts year over year. Prioritize markets for contingency planning.	Country risk screen, year-over-year trend view, scenario cost simulator
Energy Market Strategists	Analysts at trading houses, energy companies and investment firms who track supply, prices and	Link production trends and benchmark prices (Brent, WTI) to country trade positions. Identify which	Production and price trend views, net trade categorization, exporter-windfall and importer-cost

Stakeholder	Organizational Role	Decision-Support Needs	Primary Platform Outputs
	trade flows.	importers and exporters are most sensitive to each price scenario. Size the theoretical transfer between importers and exporters.	simulator

2. User Stories & Use Cases

2.1 User Story 1: Cross-Border Energy Shock Exposure

Story: As a **macroeconomic policy analyst**, I want to compare how exposed different countries are to a crude price spike, scaled to the size of each economy, so that I can identify which economies face the largest relative shock and prioritize them in my assessment.

Acceptance Criteria:

- Every country is classified as a net importer or net exporter using net crude trade (imports minus exports).
- Exposure is shown in USD and as a percentage of GDP.
- Countries with missing trade or GDP data are shown as "no data" and are not ranked as zero.

Use Case UC-1	Cross-Border Exposure Screening
Primary Actor	Macroeconomic Policy Analyst
Precondition	Data model is loaded and relationships between the country, year, trade, macro and consumption tables are active.
Main Flow	1. Analyst opens the Exposure page and selects a year. 2. Analyst filters by region and income group. 3. The dashboard classifies each country as net importer or net exporter. 4. Analyst sets the price-shock level (+10%, +25%, +50% or a custom value). 5. The dashboard recalculates extra import cost, extra export revenue and net exposure in USD and % of GDP. 6. Analyst sorts by % of GDP, drills through to a country profile, and exports the shortlist.
Alternate Flow	If a selected country has no trade or GDP value for the chosen year, it appears as "no data" with an explanatory tooltip and is excluded from rankings and totals.
Postcondition	A ranked, comparable list of exposed economies with the sample size stated on screen.

2.2 User Story 2: Currency-Adjusted Import Bill Simulation

Story: As a **supply chain risk manager**, I want to see a country's crude import bill in local currency under different price scenarios, so that I can judge how a weak exchange rate multiplies the cost of an oil spike in markets where my company operates.

Acceptance Criteria:

- Local-currency import bill equals crude imports in USD multiplied by the official annual exchange rate (LCU per USD).
- The simulation applies the selected price shock before the currency conversion.
- The currency effect and the price effect are shown separately.

Use Case UC-2	Local-Currency Import Bill Simulation
Primary Actor	Global Supply Chain Risk Manager
Precondition	Trade and exchange-rate tables are loaded for the countries of interest.
Main Flow	1. Manager selects one or more countries and a year range. 2. Dashboard shows the baseline import bill in USD and in local currency. 3. Manager sets the price-shock level. 4. Dashboard recalculates the bill in USD and LCU. 5. Dashboard decomposes the change into a price effect and a currency effect. 6. Manager compares countries and flags those where the local-currency burden grew fastest.
Alternate Flow	If an exchange-rate observation is missing for a country-year, the local-currency value is left blank and the USD value remains available.
Postcondition	A local-currency cost view that separates price effects from currency effects for each selected market.

3. Functional Requirements

ID	Requirement	Description	Acceptance Test
FR-1	**Dynamic multi-variable filtering**	The dashboard filters all pages simultaneously by country, region, income group, year range and importer/exporter status. Monthly and annual data respond to the same time filter through a shared date dimension.	A selection on any filter updates every visual on every page consistently.

ID	Requirement	Description	Acceptance Test
FR-2	**Net trade categorization and vulnerability ratios**	The SQL pipeline calculates net crude trade, flags each country as net importer or net exporter, and calculates Crude Import Burden (% of GDP), Crude Export Dependence (% of GDP) and Oil Intensity (TWh per $ billion GDP).	SQL results match the DAX measures for every country with complete inputs.
FR-3	**Scenario-based What-If price simulation**	A What-If parameter applies price shocks of +10%, +25%, +50% or a custom percentage to observed crude trade values. It returns extra import cost, extra export revenue and net exposure, in USD and as % of GDP.	Results reconcile with an independent SQL or Python calculation for a sample of countries.
FR-4	**FX-adjusted local-currency calculations**	The pipeline converts USD import values into local-currency values using official annual exchange rates, and decomposes year-over-year changes into a USD effect and a currency effect.	Local-currency values equal USD values multiplied by the rate, and the two effects sum to the total change.

4. Non-Functional Requirements

ID	Requirement	Standard	How It Is Verified
NFR-1	**Data integrity and provenance**	Every value traces back to its source file and institution. Country keys are valid ISO3 with no duplicates or orphan keys. Derived metrics are reproducible from raw data.	Key-integrity checks in SQL, row-count reconciliation after each load, and a Data Audit Log.
NFR-2	**Responsive performance**	Visual interactions respond in under one second on the full model.	Power BI Performance Analyzer on every report page.
NFR-3	**Missing-value preservation**	Missing source values are never converted to zero and never imputed. Ratios are left blank when an input is missing. Each visual shows its sample size.	Test cases that confirm blanks stay blank, and a check that no missing value was zero-filled in any table.
NFR-4	**Intuitive UI/UX navigation**	Consistent page layout, clear labels with units, tooltips that explain each metric, and a three-step flow: Overview, Analysis, Scenario.	User Acceptance Testing: a new user completes the core tasks without guidance.

5. Advanced Analytical Business Questions

#	Analytical Question	Data Used	Method
Q1	**Currency compounding.** In years of sharp benchmark price increases, how much of the rise in net importers' local-currency import bills	Oil trade, exchange rates, macro indicators, market prices	Year-over-year decomposition of the local-currency bill into price, currency and interaction components; comparison with

#	Analytical Question	Data Used	Method
	comes from the USD price-and-volume effect, how much from currency depreciation, and how much from the interaction of the two? Does the currency share align with inflation spikes?		inflation
Q2	**Structural vulnerability.** Which economies combine high oil intensity with high import burden relative to GDP, and how stable is that group when the weighting of the two dimensions changes?	Oil consumption, oil trade, macro indicators	Composite index from percentile ranks of the two ratios; sensitivity test of weights; breakdown by income group
Q3	**Benchmark price dynamics vs. trade-position shifts.** How do annual Brent and WTI price movements, and the spread between them, relate to year-over-year shifts in net crude trade across income groups, and does the spread add explanatory power beyond the price level?	Market prices (aggregated to annual), oil trade, country dimension	Annual price aggregation; correlation by income group with effective sample size reported
Q4	**Exposure scaling and rank movement.** How does exposure change between a +25% and a +50% shock when measured in absolute USD, as % of GDP and in local currency, and which top net importers move most	Oil trade, macro indicators, exchange rates	Scenario deltas under three normalizations; rank-change analysis

#	Analytical Question	Data Used	Method
	in rank between those measures?		
Q5	**Global transfer between importers and exporters.** Under a +50% shock, how large is the theoretical transfer from net importers to net exporters, and which income groups and regions are net payers or net receivers?	Oil trade, country dimension	Aggregation of net exposure by group over the complete-data sample, with CIF/FOB caveat
Q6	**Production decline and import dependence.** Does a sustained fall in a country's domestic crude production precede a rise in its crude import burden, and with what lag?	Monthly production, oil trade, macro indicators	Annualized production trend vs. import burden using lagged comparisons
Q7	**Robustness to missing data.** Which countries remain in the top-exposure group regardless of which years or sub-samples are used, and which appear only because of data coverage gaps?	All trade, macro and FX tables	Rank-stability analysis across years and complete-case subsets; list of countries whose exposure is unobservable

Analytical notes.

- The price scenarios apply to observed trade values and assume unchanged volumes. In USD terms, exposure therefore scales in proportion to the shock. Differences between the +25% and +50% cases in Q4 arise from normalization (GDP, currency) and rank changes, not from a behavioral response.
- Q3 uses annual aggregates, so each country has at most 16 yearly observations, and results are reported with their sample size.
- Imports are valued on a CIF-type basis and exports on an FOB-type basis. Q5 and any net-trade result carry this caveat.